

MATH 4720/MSSC 5720 Spring 2026 Quiz 3

Mar 31, 2026

Name: _____ **Student ID:** _____

1. A 95% confidence interval for $\mu_1 - \mu_2$ is $(1.2, 5.8)$. If we test $H_1 : \mu_1 > \mu_2$ at $\alpha = 0.05$, do we reject H_0 or not? Briefly explain.

Answer:

2. For each scenario below, choose the appropriate test: a paired t -test, Welch's t -test, or a two-sample pooled t -test.
 - (a) A teacher wants to compare the mean quiz scores of students in Section 1 and Section 2 of the same course. The two sections contain different students, and the sample standard deviations are quite different.
 - (b) A researcher measures the blood pressure of 12 patients before treatment and again after 4 weeks of treatment.
 - (c) Two independent random samples are taken from two populations. The population distributions are approximately normal, and it is reasonable to assume the population variances are equal.

Answer: